BASE24-billpay™

Billing Application Manual

Print Record

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Contents

Applied Communications, Inc.

Preface

This manual describes the BASE24-billpay Billing Application, including initial configuration requirements and daily operating procedures. This manual also provides screen illustrations and field definitions for the BASE24 CRT screens used to maintain the BASE24-billpay Billing Application database.

Audience

This manual is intended for CRT operators responsible for configuring the BASE24-billpay Billing Application, for operations personnel responsible for operating the BASE24-billpay Billing Application on a daily basis, and for personnel responsible for establishing pricing strategy.

Prerequisites

Persons using this manual should be familiar with the ***BASE24 CRT Access Manual*** before reading this manual. The ***BASE24 CRT Access Manual*** includes information about logging on to BASE24, accessing the appropriate screens, and using function keys. Persons using this manual should also be familiar with BASE24-billpay processing, including the transaction sources, types, and action codes described in the ***BASE24 Remote Banking Transaction Processing Manual.***

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** describes the screens used to maintain files and tables shared among several BASE24 products.
- The ***BASE24 Core Files and Tables Maintenance Manual*** describes the screens used to maintain files and tables used by the BASE24 Remote Banking products.
- The ***BASE24-billpay Tables Maintenance Manual*** describes the screens used to maintain tables used exclusively by the BASE24-billpay product.
- The ***BASE24 CRT Access Manual*** includes information about logging on to BASE24, accessing the appropriate screens, and using function keys.
- The ***BASE24 Remote Banking Transaction Processing Manual*** provides a list of valid action codes for the BASE24 Remote Banking products.
- The ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** provides instructions for configuring and running the Cleanup and Archive programs, which are used to remove or archive obsolete data from various BASE24-billpay tables, respectively.

Software

This manual documents standard processing for BASE24-billpay Release 1.1.

Customer-specific code modifications (that is, RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides a brief description of the BASE24-billpay Billing Application and identifies the tables specific to the BASE24-billpay Billing Application.

Section 2, "Configuration Requirements and Daily Operating Procedures," describes the BASE24-billpay Billing Application programs and presents the initial configuration requirements and daily operating procedures for using these programs.

Section 3, "Billing Group Table (BLG)," describes the screen used to maintain the BLG. The BLG defines groups of customers or financial institutions that are billed on the same cycle.

Section 4, "Billing Rate Table (BLRT)," describes the screen used to maintain the BLRT. The BLRT defines the billing rates used for each billing type and identifies the period of time the billing rate is effective.

Section 5, "Billing Type Table (BLTY)," describes the screen used to maintain the BLTY. The BLTY defines the groups of customers or financial institutions used for applying billing rates and cycles, including regular rates and special offers.

Section 6, "Rate Table (RATE)," describes the screen used to maintain the RATE. The RATE contains the information needed to identify the transactions for which a financial institution or customer is charged.

Section 7, "Rate Group Table (RATG)," describes the screen used to maintain the RATG. The RATG contains details on how to calculate the fee assessed for a transaction.

Appendix A, "Pricing Configuration Example," illustrates the procedure to use in setting up pricing information in the BLG, BLRT, BLTY, RATE, and RATG.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular column from a table.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, TB-BP300-01 for the ***BASE24-billpay Billing Application Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 04/97 for April, 1997) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a new manual. Later versions can be updates or reissues, depending on the changes being made. An update typically involves changes on a small number of pages throughout the manual. A reissue occurs when all pages of the manual are replaced because changes affect a large number of pages.

Conventions Used in this Manual

This section explains how Structured Query Language (SQL) table maintenance field descriptions and unlabeled fields are documented in this manual.

Field Descriptions

The following is an example of a field description for the Billing Rate Table (BLRT).

Example

FIID — The financial institution identifier. The value in this field links the billing information in this row to a billing group and must match an FIID that already exists in the Bank Table (BANK), which is accessed from an Institution Definition File (IDF) screen. A row with values that match the values in this field and the Billing Type field must already exist in the Billing Type Table (BLTY).

Field Length:	1–4 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLRT.FIID

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term *alphanumeric* includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term *hexadecimal* includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System-protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when function key **F8** is pressed to clear the screen.
Column Name	Provides the SQL column name associated with the field appearing on the screen.

Unlabeled Fields

Angle brackets (< >) indicate an unlabeled field on a screen or report. An unlabeled field is a field that is present on a screen or report but does not have an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen or report, the unlabeled field appears according to its place on the screen or report and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the topical index.

Applied Communications, Inc.

Section 1

Introduction

This section provides an introduction to the BASE24-billpay Billing Application, which is an add-on to the BASE24-billpay product. The section includes an overview of product functionality, an introduction to the tables specific to the BASE24-billpay Billing Application, and a discussion of the function keys used to access the tables.

BASE24-billpay Billing Application Overview

The BASE24-billpay Billing Application is an add-on to the BASE24-billpay product that offers the following features:

- A flexible pricing structure for billpay services that can be based on any combination of the following variables:
  - Transaction source, such as interactive voice response system, customer service representative terminal, or personal computer.
  - Transaction type, such as immediate payment, future payment or transfer, or recurring payment or transfer.
  - Transaction result, such as approved transactions or declined transactions.
  - Service fee assessment regardless of transaction activity.
- Ability to compute activity-based charges for billpay services used with individual customer IDs.
- Ability to accumulate daily financial institution billing statistics and compute the associated fees.
- Ability to automatically schedule customer fee payments.
- Reports that document the charges computed for individual customer IDs and financial institutions.

Section 2 describes these features in detail, including the procedures for configuring and operating the BASE24-billpay Billing Application.

Sections 3 through 7 describe the screens used to maintain the database for the BASE24-billpay Billing Application pricing structure.

Appendix A presents an example that illustrates the procedure used in setting up pricing information.

BASE24-billpay Billing Application Tables

The BASE24-billpay Billing Application uses a number of Structured Query Language (SQL) tables to configure pricing information, assign pricing information to customers and financial institutions, and calculate service charges. Tables for which table maintenance screens are provided and the BASE24 manuals in which these screens are documented are shown below. All tables documented in this manual are accessed from the Billpay Product Menu.

Bank Table (BANK)	The Bank Table (BANK) contains institution processing parameters and default values specific to the BASE24-billpay product. The BASE24-billpay Billing Application obtains vendor numbers for customer billing and the billing group and type for financial institution billing from the BANK when the Billing Application is configured for automatic fee assessment. The BANK is documented in the ***BASE24 Base Files Maintenance Manual*** because its screen is accessed from specific Institution Definition File (IDF) screens.
Billing Group Table (BLG)	The Billing Group Table (BLG) defines groups of customers or financial institutions that are billed during the same cycle. It also defines groups of customers to be processed by specific copies of the Customer Billing program and groups of financial institutions to be processed by specific copies of the Bank Billing program. The BLG is documented in section 3.
Billing Rate Table (BLRT)	The Billing Rate Table (BLRT) defines the name of the billing rates used for each of the billing types and identifies the period of time a billing rate is effective. The BLRT is documented in section 4.
Billing Type Table (BLTY)	The Billing Type Table (BLTY) defines the groups of customers or financial institutions used for applying billing rates and cycles. The BLTY is documented in section 5.

Customer Table (CSTT)	The Customer Table (CSTT) defines authorization parameters and usage accumulation information for BASE24-telebanking and BASE24-billpay customers. The BASE24-billpay Billing Application obtains billing group and type information from the CSTT. If service fees are to be automatically deducted, the BASE24-billpay Billing Application also obtains the account to which a service charge is assessed for each customer from the CSTT. The CSTT is documented in the ***BASE24 Core Files and Tables Maintenance Manual.***
Customer Billpay Table (CBPT)	The Customer Billpay Table (CBPT) contains information used for online processing of transactions, including information the BASE24-billpay Billing Application uses to calculate customer service charges. CBPT screen 1 is an inquiry-only screen that is documented in the ***BASE24-billpay Tables Maintenance Manual.***
Customer Vendor Table (CVND)	The Customer Vendor Table (CVND) identifies the vendors to which a given customer is allowed to make payments. The Customer Billing program automatically adds a vendor to this table if the program is configured to schedule payments for service charges. The CVND is documented in the ***BASE24-billpay Tables Maintenance Manual.***
History Table (HIST)	The History Table (HIST) provides the transaction information used as input by the BASE24-billpay Billing Application to calculate service charges. The Customer Billing program also writes rows to the HIST if the program is configured to schedule payments for service charges. The HIST screens are inquiry-only screens and are documented in the ***BASE24-billpay Tables Maintenance Manual.***
Personal Information Table (PIT)	The Personal Information Table (PIT) contains customer mailing information displayed on the Customer Billing Report. The PIT is documented in the ***BASE24 Core Files and Tables Maintenance Manual.***

Rate Table (RATE)	The Rate Table (RATE) contains the information needed by the BASE24-billpay Billing Application to identify the BASE24-billpay transactions for which a customer or financial institution is charged. The RATE is documented in section 6.
Rate Group Table (RATG)	The Rate Group Table (RATG) contains details on how to calculate the fee assessed for a transaction. The RATG is documented in section 7.
Vendor Table (VNDR)	The Vendor Table (VNDR) is the master vendor list for the logical network, containing one row for each vendor in the BASE24-billpay system. A row for each financial institution must be added to the VNDR if the Customer Billing program is configured to schedule payments for service charges. The VNDR is documented in the ***BASE24-billpay Tables Maintenance Manual.***

Configuring Pricing Information

The following tables contain the pricing information used by the BASE24-billpay Billing Application:

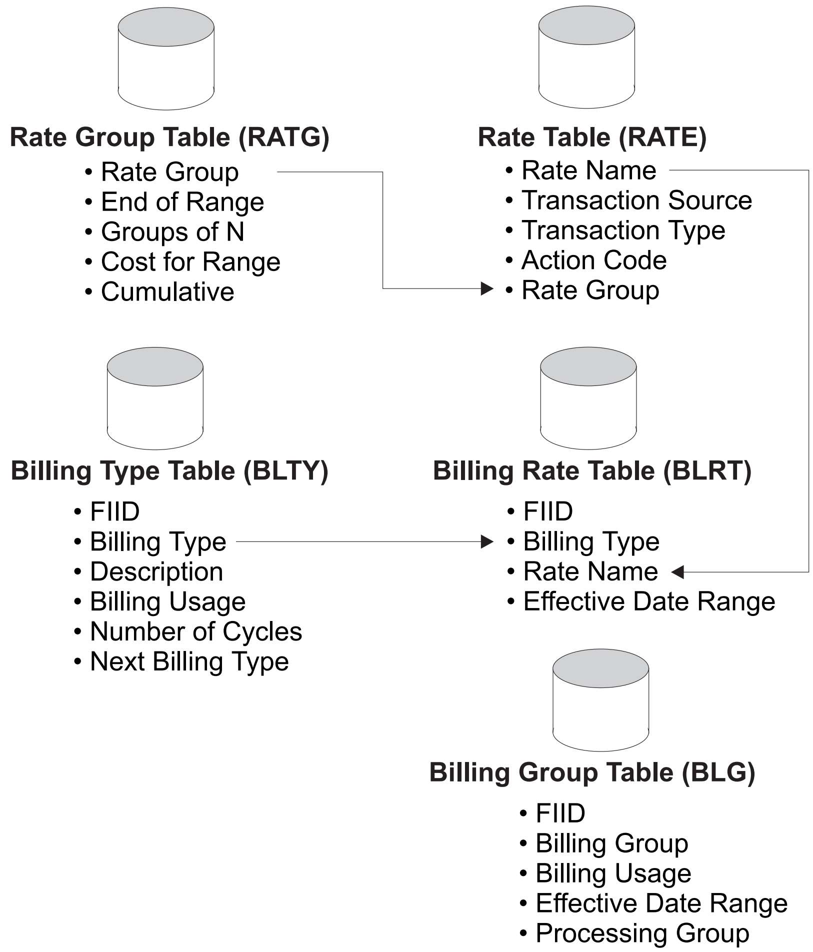
- Billing Group Table (BLG)
- Billing Rate Table (BLRT)
- Billing Type Table (BLTY)
- Rate Table (RATE)
- Rate Group Table (RATG)

The pricing information must be placed in all of these tables to establish the billing group and billing type codes that are then used in the Customer Table (CSTT) for each customer and the Bank Table (BANK) for each financial institution. Information in the tables used to establish pricing information is closely related, as shown on the following page. Each of the following values must be set up in one table before it can be used in another table:

- The rate group must be defined in the RATG before it can be used in the RATE.
- The rate name must be defined in the RATE before it can be used in the BLRT.
- The billing type must be defined in the BLTY before it can be used in the BLRT.

There are no values that must be defined in the BLG before they can be used in one of the other pricing information tables (that is, BLRT, BLTY, RATE, or RATG). However, all five of the pricing information tables must be set up to establish the billing group and billing type codes used in the CSTT and BANK.

Refer to appendix A for an example of a pricing information configuration.


Rate Group Table (RATG)
• Rate Group
• End of Range
• Groups of N
• Cost for Range
• Cumulative
Rate Table (RATE)
• Rate Name
• Transaction Source
• Transaction Type
• Action Code
• Rate Group
Billing Type Table (BLTY)
• FIID
• Billing Type
• Description
• Billing Usage
• Number of Cycles
• Next Billing Type
Billing Rate Table (BLRT)
• FIID
• Billing Type
• Rate Name
• Effective Date Range
Billing Group Table (BLG)
• FIID
• Billing Group
• Billing Usage
• Effective Date Range
• Processing Group

Assigning Pricing Information

Once pricing information has been configured, it must be assigned to each customer or financial institution. Two values are used to assign pricing information—billing group and billing type.

The billing group identifies the information needed to schedule the processing and is used to obtain information from the BLG.

The billing type identifies the information needed to calculate service charges and is used to obtain information from the BLRT, BLTY, RATE, and RATG.

If the automatic debit feature is used for customer billing, the account to be billed and the vendor (financial institution) to be paid for the service charges must also be assigned.

Customer Billing Information

Billing information for each customer is assigned in the Customer Table (CSTT) using the following fields on CSTT screen 1. The Customer Billing program uses all of these values. Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for a screen illustration and additional field descriptions for CSTT screen 1.

BILLPAY BILLING GROUP — A code identifying the billing group for this customer. This code must match a billing group defined in the BLG in order to have service charges assessed.

BILLPAY BILLING TYPE — A code identifying the billing type for this customer. This code must match a billing type in the BLRT in order to have service charges assessed.

SERVICE FEE ACCT TYPE — A code identifying the type of account to be billed for this customer’s service charges if the automatic debit feature is used for customer billing.

SERVICE FEE ACCT NUM — The account number of the account to be billed for this customer’s service charges if the automatic debit feature is used for customer billing.

Financial Institution Billing Information

Billing information for each financial institution is assigned in the Bank Table (BANK) using the following fields on BANK screen 1. The Bank Billing program uses the values in the BILLING GROUP and BILLING TYPE fields. The Customer Billing program uses the value in the VENDOR NUMBER field if the automatic debit feature is used for billing customers. Refer to the ***BASE24 Base Files Maintenance Manual*** for a screen illustration and additional field descriptions for BANK screen 1.

BILLING GROUP — A code identifying the billing group for this financial institution. This code must match a billing group defined in the BLG in order to have service charges assessed to this financial institution.

BILLING TYPE — A code identifying the billing type for this financial institution. This code must match a billing type in the BLRT in order to have service charges assessed to this financial institution.

VENDOR NUMBER — The vendor number associated with this financial institution if the automatic debit feature is used for billing customers. This vendor number must be defined in the Vendor Table (VNDR) before it can be added to the BANK.

Suggested Naming Conventions

Several of the fields displayed on the table maintenance screens used to configure the BASE24-billpay Billing Application are user-defined. Each of these fields appear in more than one table. To simplify locating the appropriate row in various tables, naming conventions should be established for the following fields:

- Billing Group
- Billing Type
- Processing Group
- Rate Group
- Rate Name

Billing Group

The billing group is a user-defined code used to divide the amount of data processed by a single Customer Billing program or Bank Billing program, allowing multiple programs to be run simultaneously. Each program can manage a portion of the data to be processed. The billing group is established in the Billing Group Table (BLG) and used in the Customer Table (CSTT) for individual customers and Bank Table (BANK) for financial institutions. One or more billing groups can be in the same processing group. However, each copy of the Customer Billing program or Bank Billing program must be in a separate processing group.

The naming convention used for the billing group should identify the group and whether the group includes individual customers or financial institutions. The code can have up to four alphanumeric characters. One example is C01 for customer (C) group 01 and B01 for bank (B) group 01. Another example is BG1C, meaning billing group (BG) 1 for customers (C) and BG1B, meaning billing group (BG) 1 for banks (B).

Billing Type

The billing type is a user-defined code that identifies a group of individual customers or financial institutions for billing purposes throughout the BASE24-billpay database. It is established in the Billing Type Table (BLTY) and used in the Billing Rate Table (BLRT) for all groups, Customer Table (CSTT) for individual customers, and Bank Table (BANK) for financial institutions.

The naming convention used for the billing type should identify all differences that could affect the fee assessed for a transaction. The code can have one or two alphanumeric characters. An example for individual customers could be RG for regular, PR for preferred regular, and SR for senior citizen regular. An example for financial institutions could be L1, L2, and L3 for different pricing levels. The billing type can also be used as part of the rate group and rate name.

Processing Group

The processing group is a user-defined code that identifies the Customer Billing program or Bank Billing program used to process data. It is established in a Logical Network Configuration File (LCONF) param and used in the Billing Group Table (BLG). One or more billing groups can be in the same processing group. However, each copy of the Customer Billing program or Bank Billing program must be in a separate processing group.

The naming convention used for the processing group should identify the group and whether the group includes individual customers or financial institutions. The code can have up to four alphanumeric characters. One example is C01 for customer (C) group 01 and B01 for bank (B) group 01. Another example is PG1C, meaning processing group (PG) 1 for customers (C) and PG1B, meaning processing group (PG) 1 for banks (B).

Rate Group

The rate group is a user-defined code that identifies a particular chargeable transaction and provides an index to the actual pricing information used to calculate the fee assessed for a transaction. It is established in the Rate Group Table (RATG) and used in the Rate Table (RATE).

The naming convention for the rate group is typically more complex than the other naming conventions shown because it should provide the information needed to identify the transaction to which it applies. The code can have up to 16 alphanumeric characters. While this code can be any user-defined value (for example, ABC), ACI recommends that you use meaningful naming conventions, such as the ones shown on the following pages.

Customer Rate Groups

One possible format for a customer rate group is *aaaa-bb-cc-dd-e*, with the following definition:

Variable	Description
aaaa	FIID of the customer financial institution. The FIID can also be used as part of the rate name.
bb	Billing type. The billing type can also be used as part of the rate name. Refer to the billing type naming conventions for examples.
cc	Transaction source. The type of device from which the transaction originates. For consistency, the code used here should match the one used in the Transaction Source field on RATE screen 1. Reserved values are as follows: AD = Audio device (for example, interactive voice response system) IB = Inbound (customer service representative terminal) PC = Personal computer SP = Screen phone
dd	Transaction type. For consistency, the code used here should match the one used in the Transaction Type field on RATE screen 1. A few of the reserved values for the field on RATE screen 1 are as follows: 5A = Do scheduled payment future 5B = Do scheduled payment recurring 9A = Schedule payment Refer to section 6 for a complete list of reserved values.
e	Approval status. Examples are A for approved and D for declined.

An example of a rate group using this format is BK02-RG-IB-9A-A, which identifies an approved (A) immediate payment transaction (9A) performed using a customer service representative terminal (IB) for a regular customer (RG) of Second Bank (BK02).

The sample customer rate group format lets you assign a unique pricing configuration for each possible customer rate group. However, this format also requires a separate RATG row for each possible customer rate group, even if some customer rate groups share the same pricing information.

You may be able to reduce the number of RATG rows you set up by eliminating unnecessary variables from the rate group format. For example, if your pricing plan does not make a distinction based on transaction source, you could set up fewer RATG rows by eliminating the transaction source from the format that you use. An example of a rate group using this format is BK02-RG-9A-A, which identifies an approved (A) immediate payment transaction (9A) for a regular customer (RG) of Second Bank (BK02).

Financial Institution Rate Groups

The key to defining financial institution rate groups is being able to identify the factors that affect the rate being charged. One possible format for a financial institution rate group is BK02-L2, which identifies Second Bank (BK02) and pricing level 2 (L2). This format can be used when all transactions are treated identically.

If a financial institution is charged one rate for transactions originating from an interactive voice response system and another rate for transactions originating from a customer service representative terminal, the rate group should be expanded to include the transaction source. The rate groups would then be BK02-L2-AD and BK02-L2-IB. Both groups identify Second Bank (BK02) and pricing level 2 (L2). However, one group is for transactions originating from an interactive voice response system (AD) and the other rate is for transactions originating from a customer service representative terminal (IB).

Pricing Impacts

The number of rate groups you use in the Rate Table (RATE) affects the way fees can be calculated. When multiple rate groups are involved, the Customer Billing and Bank Billing programs calculate the fee for each rate group and then combine the results. The same number of transactions may result in a different total fee based on the number of rate groups used. Refer to section 7 and appendix A for examples of fee calculations.

Rate Name

The rate name is a user-defined code used to organize the information needed to identify the transactions for which a fee is assessed. It is established in the Rate Table (RATE) and used in the Billing Rate Table (BLRT).

The naming convention for the rate name should identify the financial institution that uses the rate and the group of customers to which the rate applies. The code can have up to 16 alphanumeric characters. One example is BK02-RG, which identifies a regular customer (RG) of Second Bank (BK02). Another example is BK02-L2, which identifies Second Bank (BK02) and pricing level 2 (L2). The rate name can also be used as part of the rate group.

Effective Date Ranges

An effective date range for a table row controls the time frame during which the row is used, and is defined by a beginning date and an ending date. The following tables used by the BASE24-billpay Billing Application contain an effective date range:

- Billing Group Table (BLG)
- Billing Rate Table (BLRT)
- Customer Vendor Table (CVND)
- Vendor Table (VNDR)

When you validate or add a table maintenance screen for one of these tables, the beginning date defaults to the current system date and the ending date defaults to blanks, indicating a nonexpiring date range. For a nonexpiring date range, the Billpay Table Maintenance Server process automatically sets the internal database field for the ending date to be the largest available system date, which is December 31, 3999. The display of this date on table maintenance screens, customer service representative (CSR) screens, and reports is left blank to indicate that it is nonexpiring. You can always enter a specific ending date to make a row expire.

The effective date range serves a slightly different role in each table. The ending date in the BLG identifies which row should be used during processing on a particular business date. For this reason, you should always establish beginning and ending dates for each BLG row that you enter. The effective date range in the BLRT lets you establish a date to review your pricing for billing services, and you should use it for this purpose.

The effective date ranges in the CVND and VNDR, on the other hand, are used to control whether a vendor is available for use. The default values should be acceptable for dates in both of these tables unless some type of status change makes a vendor unusable.

The BASE24-billpay product includes two database management programs—Cleanup and Archive—to delete or archive expired database rows on a daily, monthly, or annual basis. Refer to the ***BASE24-billpay Tables Maintenance Manual*** and the ***BASE24-billpay End-of-Period Processing and Reporting Manual*** for additional information on the use of these programs.

Function Keys

The standard BASE24 function keys are used on BASE24-billpay screens. These function keys are described in the ***BASE24 Base Files Maintenance Manual*** and ***BASE24 CRT Access Manual***. When the use of these keys differs for a BASE24-billpay screen, the differences are noted in the section of the manual dealing with the screen grouping to which they apply.

BASE24 also supplies an online Help screen which lists the function keys that apply on each screen. Press the **F12** key to display the Help screen.

Note: Other BASE24 products use the **F13** key to change to another logical network. This key is defined for the same purpose on BASE24-billpay table maintenance screens. However, pressing the **F13** key results in an error message because BASE24-billpay Release 1.1 does not currently support multiple logical networks.

Section 2

Configuration Requirements and Daily Operating Procedures

The BASE24-billpay Billing Application gives a service provider the capability to bill customers and client financial institutions for using the BASE24-billpay product.

This section describes the BASE24-billpay Billing Application programs and presents the initial configuration requirements and daily operating procedures for using these programs.

Billing Application Processing

The BASE24-billpay Billing Application uses the following programs to accumulate and process billing information:

- Customer Billing
- Bank Day
- Bank Billing

The Customer Billing program performs all processing related to customer account billing services, including the collecting of data and the calculating of service fees, as well as the scheduling of payments for the fees, if so configured. The Bank Day and Bank Billing programs work together to perform the processing related to financial institution billing services. The Bank Day program collects data on a daily basis and the Bank Billing program calculates the service fees based on the collected data. Financial institution billing services do not include the scheduling of payments for the fees.

The Customer Billing program can be used with or without the Bank Day and Bank Billing programs. Likewise, the Bank Day and Bank Billing programs can be used with or without the Customer Billing program.

Customer Billing Program

The Customer Billing program, which is named CUSTBIL, computes the fee to be charged to each customer for the current cycle and writes the results to the Customer Billing Summary Table (CBLS). Optionally, the Customer Billing program can schedule the fee payments to be made from customer accounts.

Normal Processing

A batch scheduler such as the Tandem NetBatch product starts the Customer Billing program every day after logical network cutover time plus a lag time. Refer to the LGNT-CUTOVER and BP-LAG-TIME params in the Logical Network Configuration File (LCONF) to determine the logical network cutover time and amount of lag time, respectively. The program can also be started manually by an operator. When the Customer Billing program starts, it reads the Billing Group Table (BLG) to find the billing groups within its processing group that have the appropriate cycle end date. The cycle end date defaults to the day prior to the current business day or can be entered manually by an operator.

The Customer Billing program calculates the fee for each customer in the specified billing groups. It reads the Customer Billpay Table (CBPT), Customer Table (CSTT), and History Table (HIST) to build a table of transaction counters in internal memory, and then increments transaction counters in this table based on the transaction source, transaction type, and action code in each HIST row. The value in the Last Billing Reference Number field on CBPT screen 1 for each customer identifies the most recent HIST row used for fee calculation purposes. After writing the total fee amount to the CBLS, the Customer Billing program sets the value in the Last Billing Reference Number field on CBPT screen 1 and subtracts one from the value in the Remaining Number Of Cycles field on CBPT screen 1.

After updating the value in the Remaining Number Of Cycles field on CBPT screen 1, the Customer Billing program determines whether any changes must be made to the billing group or billing type in the CBPT prior to the next statement cycle. One check is to determine whether an automatic change should be made from the current billing type to a new billing type based on information configured in the Billing Type Table (BLTY). The second check is to determine whether the billing type or billing group have been changed in the Customer Table (CSTT). CBPT changes resulting from manual changes in the CSTT override automatic CBPT changes based on information in the BLTY. Any CBPT changes resulting from either of these checks take effect with the next billing cycle.

The need for an automatic change in the billing type based on information in the BLTY is based on the value in the Remaining Number Of Cycles field on CBPT screen 1. If the value in this field is set to zero during fee calculation, the Customer Billing program makes the following updates:

- Sets the value in the Current Billing Type field on CBPT screen 1 to the value in the Next Billing Type field on BLTY screen 1 for the current billing type.
- Sets the value in the Remaining Number Of Cycles field on CBPT screen 1 to the value in the Number Of Cycles field on BLTY screen 1 for the new billing type.

The Customer Billing program checks the CSTT for any manual changes in the billing type or billing group, as follows:

- Compares the value in the BILLPAY BILLING TYPE field on CSTT screen 1 with the value in the Original Billing Type field on CBPT screen 1. If they are different, information from fields on BLTY screen 1 for the billing type specified on CSTT screen 1 is used to update values in corresponding fields on CBPT screen 1.

- Compares the value in the BILLPAY BILLING GROUP field on CSTT screen 1 with the value in the Billing Group field on CBPT screen 1. If they are different, the value in the Billing Group field on CBPT screen 1 is replaced with the value from the BILLPAY BILLING GROUP field on CSTT screen 1.

Generating Future Payments

Optionally, the Customer Billing program can schedule future payments for customer fees, based upon the value of the BP-GENERATE-FUTURE-PMT param in the LCONF.

If this feature is enabled, all financial institutions must be configured as vendors in the Vendor Table (VNDR). You must add these vendor numbers to the VENDOR NUMBER field on Bank Table (BANK) screen 1 for each institution. The Customer Billing program uses the value in the CUSTOMER ID field on CSTT screen 1 as the account with vendor for the payment. If an appropriate entry is not already in the Customer Vendor Table (CVND), the Customer Billing program adds one.

The Customer Billing program schedules a payment by adding one row to the Future Table (FUTR) and one row to the HIST. The date of this payment is set to the day immediately following the cycle end date.

Restart and Error Processing

At startup, the Customer Billing program determines the starting point within the Billing Group Table (BLG) and the Customer Billpay Table (CBPT) by reading the Customer Billing Summary Table (CBLS). If the program stops and needs to be restarted, and the previous business day is the one being processed, you can restart the program with the default cycle end date. Otherwise, you can restart the program with a parameter that indicates the cycle end date being processed.

When an error occurs during customer billing, the Customer Billing program logs a message detailing the problem and stops. After correcting the problem, you can restart the Customer Billing program at the point where the error occurred.

TMF Considerations

All the processing for a single customer is done within the same TMF transaction. The Customer Billing program also can batch the processing for multiple customers within one TMF transaction, according to settings of the BP-TMF-BATCH-DURATION-TIME and BP-TMF-BATCH-SIZE params in the LCONF.

Multiple Program Copies

More than one copy of the Customer Billing program can be run simultaneously by defining multiple billing processing groups in the BLG.

Bank Day Program

The Bank Day program, which is named BANKDAY, accumulates bank billing statistics for the transactions that occurred on the previous business day and writes the totals for the day to the Bank Transaction Summary Table (BKTS). The Bank Day program helps spread processing resources across a broad period of time; however, it does not calculate any charges.

Normal Processing

A batch scheduler such as the Tandem NetBatch product starts the Bank Day program every day after logical network cutover plus a lag time. Refer to the LGNT-CUTOVER and BP-LAG-TIME params in the LCONF to determine the logical network cutover time and amount of lag time, respectively. The program can also be started manually by an operator.

When the Bank Day program starts, it reads the History Table (HIST) by the timestamp in the hist_gmt column of each row to count all transactions in a period of approximately 24 hours. The Bank Day program determines the start of the processing time range by selecting the last row in the Bank Transaction Summary Table (BKTS) and determines the end of the processing time range by adding the lag time to the logical network cutover time. The ending date defaults to the day prior to the current business day or can be entered manually by an operator.

Each HIST row is summarized and stored in internal memory. Each transaction is counted as part of a group. Each group is unique by values in the bus_lct, fiid, txn_src, txn_typ, and rsp_code columns of the HIST. After all transactions have been grouped, the Bank Day program inserts a row into the BKTS for each group.

Restart and Error Processing

The Bank Day program stops and logs a message if an error occurs during processing. After the error is corrected and the program is restarted, the Bank Day program returns to the original starting point for the day.

Multiple Program Copies

Multiple copies of the Bank Day program can be run simultaneously by defining one copy of the program for each partition of the HIST. The name of the Bank Day program for each HIST partition is identified by a BP-HISTORY-PARTITION-NUM param in the LCONF.

Bank Billing Program

The Bank Billing program, which is named BANKBIL, computes the fee that should be charged to each financial institution for the current billing cycle. The Bank Billing program reads the daily totals written to the Bank Transaction Summary Table (BKTS) by the Bank Day program and writes the results to the Bank Billing Summary Table (BBLS).

Normal Processing

A batch scheduler such as the Tandem NetBatch product starts the Bank Billing program every day after the Bank Day program finishes updating the BKTS. The Bank Day program runs every day after logical network cutover plus a lag time. Refer to the LGNT-CUTOVER and BP-LAG-TIME params in the LCONF to determine the logical network cutover time and amount of lag time, respectively. The program can also be started manually by an operator.

When the Bank Billing program starts, it reads the Billing Group Table (BLG) to determine the financial institutions in its processing group that have the appropriate cycle end date. The cycle end date defaults to the day prior to the current business day or can be entered manually by an operator. The Bank Billing program reads the BKTS to generate an internal table of counts indexed by transaction source, transaction type, and action code. It then calculates a fee and writes it to the BBLS.

When the Bank Billing program starts, it obtains the values in the BILLING GROUP and BILLING TYPE fields on Bank Table (BANK) screen 1 for each financial institution. Unlike the Customer Billing program, the Bank Billing program does not automatically change from one billing type to another based on information in the Number Of Cycles or Next Billing Type fields on Billing Type Table (BLTY) screen 1.

Restart and Error Processing

The Bank Billing program does not perform any special restart processing. However, it can be rerun for a given cycle end date because it ignores duplicate record errors on the BKTS.

TMF Considerations

All the processing for a single financial institution is done within the same TMF transaction. The Bank Billing program also can batch the processing for multiple financial institutions within one TMF transaction, according to settings of the BP-TMF-BATCH-DURATION-TIME and BP-TMF-BATCH-SIZE params in the LCONF.

Multiple Program Copies

More than one copy of the Bank Billing program can be run simultaneously by defining multiple processing groups in the BLG.

Billing Application Assigns and Params

The BASE24-billpay Billing Application programs use the following Logical Network Configuration File (LCONF) assigns and params. Refer to the ***BASE24 Logical Network Configuration File Manual*** for additional information about these assigns and params, including the use of the Process Name field on the LCONF Param screen for all params except LGNT-CUTOVER and OMF-AUDIT.

Assigns

The BASE24-billpay Billing Application programs use the following Logical Network Configuration File (LCONF) assign:

OMF — The fully qualified file name of the Online Maintenance File (OMF). All three programs require this assign. However, this assign is used only by the Customer Billing program when it schedules fee payments.

Params

One or more of the BASE24-billpay Billing Application programs use each of the following Logical Network Configuration File (LCONF) params:

BP-BILL-PROCESSING-GRP — The name of the processing group that a Customer Billing or Bank Billing program is to process. The value of this param must match the value in the Processing Group field on Billing Group Table (BLG) screen 1. This param must identify the symbolic process name of the Customer Billing or Bank Billing program.

BP-BILL-TEXT — The text that the Customer Billing program writes to the Customer Billing Summary Table (CBLS) and Future Table (FUTR). The default text is “Charge for BillPay.” This param uses a process name of CUST-BILLING.

BP-GENERATE-FUTURE-PMT — A code indicating whether the Customer Billing program should schedule a payment for the amount of the service fee. This param uses a process name of CUST-BILLING. The valid values are as follows:

Y = Yes, schedule a payment for the amount of the service fee.
N = No, do not schedule a payment for the amount of the service fee. Default.

BP-HISTORY-PARTITION-NUM — The partition of the History Table (HIST) processed by a specific BASE24-billpay process. This param must identify the symbolic process name of each Bank Day program.

BP-LAG-TIME — The length of time that must elapse after logical network cutover before the Remittance Generator process begins processing transactions for the next remittance business day. The Bank Day program uses this param to define the 24-hour block of History Table (HIST) rows used to update the Bank Transaction Summary Table (BKTS).

BP-NUM-FUTURE-PART-INS — The maximum number of partitions that can be inserted into the Future Table (FUTR). The Customer Billing process uses this param when it schedules fee payments. This param uses a process name of BILLPAY.

BP-NUM-HISTORY-PART-INS — The maximum number of partitions that can be inserted into the History Table (HIST). The Customer Billing process uses this param when it schedules fee payments. This param uses a process name of BILLPAY.

BP-TMF-BATCH-DURATION-TIME — The maximum length of time a Transaction Monitoring Facility (TMF) batch can exist before it is committed. The Customer Billing program uses this param with a process name of CUST-BILLING and the Bank Billing program uses this param with a process name of BANK-BILLING.

BP-TMF-BATCH-SIZE — The maximum number of rows from all tables and records from all files to be included in the same Transaction Monitoring Facility (TMF) batch. The Customer Billing program uses this param with a process name of CUST-BILLING and the Bank Billing program uses this param with a process name of BANK-BILLING.

LGNT-CUTOVER — The logical network cutover time. All three programs require this param.

OMF-AUDIT — A code controlling the extent of row logging to the Online Maintenance File (OMF). All three programs require this param. However, this param is used only by the Customer Billing program when it schedules fee payments.

Sample LCONF Configuration

The following tables provide sample LCONF entries for the BASE24-billpay Billing Application programs. This sample configuration assumes the History Table (HIST) and Future Table (FUTR) have three partitions.

Customer Billing Program

The following tables illustrate the LCONF entries for the Customer Billing program.

Process Name	Assign	Value
****************	OMF*	\B24.$DATA.PRO1DATA.OMF (data file name) \B24.$DATA.PRO1TPLT.AYYMMDDX (template file name)

Process Name	Param	Value
P1A^CUSTBL^1	BP-BILL-PROCESSING-GRP	PG1C
P1A^CUSTBL^2	BP-BILL-PROCESSING-GRP	PG2C
CUST-BILLING	BP-BILL-TEXT	Charge for EasyPay
CUST-BILLING	BP-GENERATE-FUTURE-PMT	Y
BILLPAY	BP-NUM-FUTURE-PART-INS*	3
BILLPAY	BP-NUM-HISTORY-PART-INS*	3
CUST-BILLING	BP-TMF-BATCH-DURATION-TIME	60
CUST-BILLING	BP-TMF-BATCH-SIZE	100
****************	LGNT-CUTOVER*	19:00
****************	OMF-AUDIT*	1

* This assign or param should already be configured in the LCONF for use by other BASE24-billpay processes.

Bank Day and Bank Billing Programs

The following tables provide sample LCONF entries for the Bank Day and Bank Billing programs.

Process Name	Assign	Value
****************	OMF*	\B24.$DATA.PRO1DATA.OMF (data file name) \B24.$DATA.PRO1TPLT.AYYMMDDX (template file name)

Process Name	Param	Value
P1A^BANKBL^1	BP-BILL-PROCESSING-GRP	PG1B
P1A^BANKBL^DAY^1	BP-HISTORY-PARTITION-NUM	1
P1A^BANKBL^DAY^2	BP-HISTORY-PARTITION-NUM	2
P1A^BANKBL^DAY^3	BP-HISTORY-PARTITION-NUM	3
BILLPAY	BP-LAG-TIME*	15:00
BANK-BILLING	BP-TMF-BATCH-DURATION-TIME	60
BANK-BILLING	BP-TMF-BATCH-SIZE	100
****************	LGNT-CUTOVER*	19:00
****************	OMF-AUDIT*	1

* This assign or param should already be configured in the LCONF for use by other BASE24-billpay processes.

Running Billing Application Programs

All three BASE24-billpay Billing Application programs are batch jobs that can be scheduled to run each day using a batch scheduler such as the Tandem NetBatch product or can be started manually. Either way, the defines for the BASE24-billpay environment must be set so that the define names can be resolved to the correct physical file and table locations prior to running any of the BASE24-billpay Billing Application programs. The ENVSET TACL macro is used to set the environment.

Each program can be started by running the appropriate TACL macro (GOCUSTBL, GOBANKDY, or GOBANKBL). These TACL macros include the ENVSET TACL macro and the parameters for running each program.

Customer Billing Program

The Customer Billing program can be started from a TACL prompt using the GOCUSTBL macro.

Macro Example

The following GOCUSTBL macro example assumes two processing groups are being used:

```
?tacl macro
envset
== Customer Billing for the first processing group
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].custbil  $B.PRO1EXEC.L1CONF   P1A^CUSTBL^1
==
== Customer Billing for the second processing group
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].custbil  $B.PRO1EXEC.L1CONF   P1A^CUSTBL^2
```

TACL Command

The GOCUSTBL macro is based on the following TACL command:

Syntax

CUSTBIL *lconf sym-process-name* [*cycle-end-date*] [VERIFY] *lconf* — The name of the LCONF that contains the assign and params needed for the program to run.

sym-process-name — The symbolic process name of a specific Customer Billing program. This value is used to look up the BP-BILL-PROCESSING-GRP param in the LCONF and identify the processing group a particular Customer Billing program is to process.

cycle-end-date — The ending date (YYYYMMDD) of the billing cycle being processed. Transactions will be processed up to the logical network cutover time plus lag time on this date.

The default value for this parameter is the current business day minus one day. For example, if this program is started after the logical network cutover time plus lag time on December 1, the business day would be December 2 and the ending date would be December 1.

VERIFY — When this optional parameter is included in the command, the Customer Billing program verifies the configuration files it uses without processing any transactions.

Bank Day Program

The Bank Day program can be started from a TACL prompt using the GOBANKDY macro.

Macro Example

The following GOBANKDY macro example assumes the History Table (HIST) has three partitions:

```
?tacl macro
envset
== Daily Bank Billing processing for the first HIST partition
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].bankday  $B.PRO1EXEC.L1CONF   P1A^BANKBL^DAY^1
==
== Daily Bank Billing processing for the second HIST partition
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].bankday  $B.PRO1EXEC.L1CONF   P1A^BANKBL^DAY^2
==
== Daily Bank Billing processing for the third HIST partition
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].bankday  $B.PRO1EXEC.L1CONF   P1A^BANKBL^DAY^3
```

TACL Command

The GOBANKDY macro is based on the following TACL command:

Syntax

BANKDAY *lconf sym-process-name* [*run-date*]

lconf — The name of the LCONF that contains the assign and params needed for the program to run.

sym-process-name — The symbolic process name of a specific Bank Day program. This value is used to look up the BP-HIST-PARTITION-NUM param in the LCONF and identify the History Table (HIST) partition a particular Bank Day program is to read.

run-date — The business date (YYYYMMDD) of the transactions being processed. Transactions will be processed up to the logical network cutover time plus lag time on this date.

The default value for this parameter is the current business day minus one day. For example, if this program is started after the logical network cutover time plus lag time on December 1, the business day would be December 2 and the ending date would be December 1.

Bank Billing Program

The Bank Billing program can be started from a TACL prompt using the GOBANKBL macro.

Macro Example

The following GOBANKBL macro example assumes two processing groups are being used:

```
?tacl macro
envset
== Bank Billing for the first processing group
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].bankbil  $B.PRO1EXEC.L1CONF   P1A^BANKBL^1
==
== Bank Billing for the second processing group
==                        <LCONF File Name>  <Sym Proc Name>
[:bp_obj_subvol].bankbil  $B.PRO1EXEC.L1CONF   P1A^BANKBL^2
```

TACL Command

The GOBANKBL macro is based on the following TACL command:

Syntax

BANKBIL *lconf sym-process-name* [*cycle-end-date*]

lconf — The name of the LCONF that contains the assign and params needed for the program to run.

sym-process-name — The symbolic process name of a specific Bank Billing program. This value is used to look up the BP-BILL-PROCESSING-GRP param in the LCONF and identify the processing group a particular Bank Billing program is to process.

cycle-end-date — The ending date (YYYYMMDD) of the billing cycle being processed. Transactions will be processed up to the logical network cutover time plus lag time on this date.

The default value for this parameter is the current business day minus one day. For example, if this program is started after the logical network cutover time plus lag time on December 1, the business day would be December 2 and the ending date would be December 1.

Billing Application Reports

The BASE24-billpay Billing Application uses TACL macros and routines with the SQL conversational interface (SQLCI) report writer to generate the following reports:

- Customer Billing Report
- Bank Billing Report

Customer Billing Report

The RPCBLS report routine uses the SQLCI report writer to generate the Customer Billing Report from information placed in the Customer Billing Summary Table (CBLS) by the Customer Billing program, as well as customer mailing information from the Personal Information Table (PIT).

A sample of the Customer Billing Report is shown on the next page, followed by field descriptions. The fields are repeated for each CBLS entry.

```
<customer name>                            <customer id>                                        <date-fmt>
<address line 1>
<address line 2>
<city>                   <st>  <postalcde>


                    Monthly Billpay Billing for Cycle beginning <begin-date> Ending <end-date>

<billing text>                                                                              <svc-charge>



<customer name>                            <customer id>                                        <date-fmt>
<address line 1>
<address line 2>
<city>                   <st>  <postalcde>


                    Monthly Billpay Billing for Cycle beginning <begin-date> Ending <end-date>

<billing text>                                                                              <svc-charge>



<customer name>                            <customer id>                                        <date-fmt>
<address line 1>
<address line 2>
<city>                   <st>  <postalcde>


                    Monthly Billpay Billing for Cycle beginning <begin-date> Ending <end-date>

<billing text>                                                                              <svc-charge>



<customer name>                            <customer id>                                        <date-fmt>
<address line 1>
<address line 2>
<city>                   <st>  <postalcde>


                    Monthly Billpay Billing for Cycle beginning <begin-date> Ending <end-date>

<billing text>                                                                              <svc-charge>
```

<customer name> — The given name or first name of this customer, followed by the customer's middle initial and the family name or last name of this customer.

<customer id> — The customer ID of a Customer Table (CSTT) row this customer is allowed to access.

<date-fmt> — The date this report was generated, displayed in the format specified by the BP-DATE-FORMAT param in the LCONF.

<address line 1> — The first line of the street or mailing address for this customer.

<address line 2> — The second line of the street or mailing address for this customer.

<city> — The name of the city for this customer's residence or mailing address.

<st> — The two-letter U.S. state or Canadian province abbreviation for this customer's residence or mailing address.

The U.S. state codes are based on the 1988 ANSI X3.38 standard, ***Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange***. However, a value other than the ANSI code can be used.

<postalcde> — The postal code of this customer's residence or mailing address.

<begin-date> — The beginning date of the billing cycle for which this service charge applies.

<end-date> — The ending date of the billing cycle for which this service charge applies.

<billing text> — The Billpay service description written to the CBLS. This description is also written to the Future Table (FUTR) if customers are billed for the Billpay service.

<svc-charge> — The amount of the service charge, expressed in whole and fractional currency units, computed for this customer.

Bank Billing Report

The RPBBLS report routine uses the SQLCI report writer to generate the Bank Billing Report from information placed in the Bank Billing Summary Table (BBLS) by the Bank Billing program. The heading of the report contains the ending date of the billing cycle for which the report is generated.

A sample of the report is shown on the next page, followed by field descriptions.

```
                      Bank Billing Report for Cycle Ending on 970103

                            Tran    Tran  Response
FIID  Rate Group            Source  Type  Code      Rate Count   Total Amount
----  ----------------      ------  ----  --------  -----------  ------------

BK02  BK02-L2-IB            AD      9A    000            24,000
                                                    -----------
      Subtotal:                                          24,000     $2,400.00


      BK02-L2-AD            IB      5A    000               200
                            IB      5B    000               300
                            IB      9A    000             6,000
                                                    -----------
      Subtotal:                                           6,500     $1,300.00



                                                    -----------  ------------
      FIID Total:                                        30,500     $3,700.00
```

FIID — The financial institution for which the transaction counts and service charge amounts were computed.

The entry labeled FIID Total shows the total number of transactions for this FIID in the Rate Count column and the total service charge amount for this FIID in the Total Amount column.

Rate Group — The name of the rate group for which the transaction counts and service charge amounts were computed. The rate group identifies the pricing information used to compute the service charges.

The entry labeled Subtotal shows the total number of transactions for this rate group in the Rate Count column and the total service charge amount for this rate group in the Total Amount column.

Tran Source — A code that identifies the type of device from which the transaction originates. These codes can be user-defined or one of the following reserved values:

AD = Audio device (for example, interactive voice response system)
BL = Billing application
IB = Inbound (customer service representative terminal)
PC = Personal computer
SP = Screen phone

Tran Type — A code that identifies the transaction type. The code may be a reserved value or be user-defined. Refer to section 6 for a complete list of reserved values.

Response Code — A code used by the transaction authorizer to indicate the processing outcome of a transaction. Refer to section 6 for a list of valid response codes, also known as action codes.

Rate Count — The number of transactions that occurred during the billing cycle with the combination of rate group, transaction source, transaction type, and response code shown on this line of the report.

Total Amount — The total service charge assessed for the number of transactions counted for a rate group or FIID. The subtotal charge for each rate group is based on the total number of transactions for the rate group, not the number of transactions for each combination of transaction source, transaction type, and response code within the rate group. The FIID total charge is the sum of the subtotal charges for the rate groups.

Running Billing Application Reports

The BASE24-billpay Billing Application uses TACL macros and routines with the SQL conversational interface (SQLCI) report writer to generate the Customer Billing Report and Bank Billing Report. Both reports can be scheduled to run each day using a batch scheduler such as the Tandem NetBatch product or can be started manually. Either way, the following configuration information must be available before the reports can be generated:

- The defines for the BASE24-billpay environment must be set so that the define names can be resolved to the correct physical file and table locations. The ENVSET TACL macro is used to set the environment.
- TACL Process Definition commands must be available for use by the RPCBLS and RPBBLS report routines. The :UTILS:DP value is used in a TACL USE command or TACL use list to add the TACL Process Definition commands.

Each report routine can be started by running the appropriate TACL macro (GORPCBLS or GORPBBLS). These TACL macros include the ENVSET TACL macro and parameters for running each program. However, the TACL Process Definition commands must be available before these TACL macros are run.

Customer Billing Report

The Customer Billing Report can be started from a TACL prompt using the GORPCBLS macro. The spooler location for the Customer Billing Report is $S.#BP.CBLS.

Macro Example

The following GORPCBLS macro example assumes that the Customer/Personal ID Relation Table (CPIT) is available:

```
?tacl macro
envset !
==                     <Date Format> <CPIT Present> <Run Date>
==
[:bp_rpts_subvol].rpcbls   M2D2Y2           Y        1997-01-15
```

TACL Command

The GORPCBLS macro is based on the following TACL command:

Syntax

RPCBLS *date-format cpit-present cycle-end-date*

RPCBLS — The name of the TACL routine for the Customer Billing Report.

date-format — A code identifying the format used to display all dates on the report. Valid values are as follows:

D2M2Y2 = DD MM YY (for example, 03 01 97)
D2M2Y4 = DD MM YYYY (for example, 03 01 1997)
D2M3Y2 = DD MMM YY (for example, 03 JAN 97)
M2D2Y2 = MM DD YY (for example, 01 03 97)
M2D2Y4 = MM DD YYYY (for example, 01 03 1997)
M3D2Y2 = MMM DD YY (for example, JAN 03 97)
M3D2Y4 = MMM DD YYYY (for example, JAN 03 1997)

cpit-present — A code that identifies whether the Customer/Personal Information Relation Table (CPIT) is supported. The value of this parameter determines how PIT row information is selected for inclusion in the report. Valid values are as follows:

Y = Yes, the CPIT is supported.
N = No, the CPIT is not supported.

When the CPIT is not supported, the PIT row used is determined by matching the customer ID in the Customer Billing Summary Table (CBLS) with the personal ID in the PIT.

When the CPIT is supported, the PIT row used is determined by matching the customer ID in the CBLS with the primary account number in the CPIT, then matching the personal ID in the CPIT row with the personal ID in the PIT.

cycle-end-date — The ending date (YYYY-MM-DD) of the billing cycle being reported.

Bank Billing Report

The Bank Billing Report can be started from a TACL prompt using the GORPBBLS macro. The spooler location for the Bank Billing Report is $S.#BP.BBLS.

Macro Example

The following GORPBBLS macro example uses a cycle-end date of January 15, 1997:

```
?tacl macro
envset !
==                       <Run Date>
==
[:bp_rpts_subvol].rpcbls  1997-01-15
```

TACL Command

The GORPBBLS macro is based on the following TACL command:

Syntax

RPBBLS cycle-end-date

RPBBLS — The name of the TACL routine for the Bank Billing Report.

cycle-end-date — The ending date (YYYY-MM-DD) of the billing cycle being processed.

Cleanup Program

Cleanup is a batch program run after logical network cutover on a daily basis to remove obsolete data from BASE24-billpay tables. Preferably, the Cleanup program should be run as one of the last steps in the daily processing cycle.

The Cleanup program can remove obsolete rows from a number of BASE24-billpay tables, including the following tables used by the BASE24-billpay Billing Application:

- Bank Billing Summary Table (BBLS)
- Bank Transaction Summary Table (BLTS)
- Billing Group Table (BLG)
- Billing Rate Table (BLRT)
- Customer Billing Summary Table (CBLS)

Obsolete data should be removed from the BBLS, BLTS, and CBLS daily and should be removed from the BLG and BLRT annually. Refer to the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual*** for instructions on configuring and running the Cleanup program, including suggested values for the program parameters associated with each table.

Section 3

Billing Group Table (BLG)

The Billing Group Table (BLG) defines groups of customers or financial institutions that are billed on the same cycle. Each customer or financial institution using the BASE24-billpay Billing Application must be assigned to a billing group.

A BLG row is required for each billing cycle that occurs. For example, a monthly billing cycle requires 12 BLG rows per year and a bi-monthly billing cycle requires 6 BLG rows per year.

The BLG also provides the information needed for processing groups, enabling a billing group to be processed concurrently by multiple copies of the Customer Billing program or Bank Billing program. Each copy of the Customer Billing or Bank Billing program requires a separate set of BLG rows. For example, if three copies of the Customer Billing program are used, a monthly billing cycle requires 36 BLG rows per year and a bi-monthly billing cycle requires 18 BLG rows per year.

The key to the BLG is a combination of the values in the FIID, Billing Group, Billing Usage, and Cycle End Date fields. This key must be unique when a row is added and cannot be changed when a row is updated.

The value in the Billing Group field on BLG screen 1 is used on file maintenance screens for several other tables. Therefore, a BLG row should be available as long as that value is used in any of the following fields or columns:

- The BILLING GROUP field on Bank Table (BANK) screen 1
- The BILLPAY BILLING GROUP field on Customer Table (CSTT) screen 1
- The Billing Group field on Customer Billpay Table (CBPT) screen 1
- The bill_grp column in the Customer Billing Summary Table (CBLS)

The Billpay Table Maintenance Server process does not stop you from deleting a BLG row that is being used in one of these tables. It is up to you to ensure that the BLG row is no longer needed.

Screen 1

BLG screen 1 contains general information for each of the billing groups defined for an FIID and statement cycle. BLG screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  BILLING GROUP         LLLL        YY/MM/DD  HH:MM    1 OF  1

                  FIID ____

         Billing Group ____

         Billing Usage _ (B=Bank, C=Customer)

 Effective Date Range:
      Cycle Begin Date ________ yyyymmdd
        Cycle End Date ________

      Processing Group ______





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```

FIID — The financial institution identifier used for the billing group.

For bank billing, the value in this field links the billing group to the financial institution and must match an FIID that already exists in the Bank Table (BANK), which is accessed from Institution Definition File (IDF) screens.

For customer billing, the value in this field must be hyphens (----), indicating a single billing group.

Field Length:	1–4 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLG.FIID

Billing Group — A code identifying the billing group, which is an arbitrary value used to balance the number of customers or financial institutions being processed on a particular date.

Refer to the "Suggested Naming Conventions" topic in section 1 for naming the billing group.

Field Length:	1–4 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLG.BILL_GRP

Billing Usage — A code identifying the type of billing group in this row. Valid values are as follows:

B = Bank billing. The value in the FIID field must be in the Bank Table (BANK).
C = Customer billing. The value in the FIID field must be hyphens (----).

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	No default value
Column Name:	BLG.BILL_USAGE

Effective Date Range

The dates during which this billing cycle is applicable. Billing cycles for the same combination of FIID, billing group, and billing usage cannot overlap. A separate BLG row is required for each billing cycle because the Customer Billing program and Bank Billing program use the date in the Cycle End Date field to determine which billing groups to process on a given business date.

The format used to enter and display the dates in these fields is shown to the right of the Cycle Begin Date field and is controlled by the BP-DATE-FORMAT param in the Logical Network Configuration File (LCONF).

Cycle Begin Date — The start of the billing cycle.

If the field is left blank, the system fills in the current date when the row is added or updated. However, the correct start date for each billing cycle should be entered so that billing cycles do not overlap.

Field Length: 8 numeric characters
Required Field: No
Default Value: No default value
Column Name: BLG.CYCLE_BEG_DATE

Cycle End Date — The end of the billing cycle.

This field must contain the correct ending date for each billing cycle because the Customer Billing program and Bank Billing program use the date in this field to determine which billing groups to process on a given business date.

If this field is left blank when the row is added or updated, the system assigns a date of December 31, 3999, but does not display it.

Field Length: 8 numeric characters
Required Field: No
Default Value: No default value
Column Name: BLG.CYCLE_END_DATE

Processing Group — A value that identifies the Customer Billing program or Bank Billing program that uses the information in this BLG row.

This value is used to break up the amount of data processed by a single Customer Billing program or Bank Billing program. One or more billing groups can be in the same processing group. This allows multiple Customer Billing programs or Bank Billing programs to be run, each managing a portion of the data to be processed.

Each copy of the Customer Billing program or Bank Billing program has an LCONF BP-BILL-PROCESSING-GROUP param it uses to access the BLG. The value in this field must match the value of the BP-BILL-PROCESSING-GROUP param for each copy.

Refer to the "Suggested Naming Conventions" topic in section 1 for naming the processing group.

Field Length: 1–6 alphanumeric characters
Required Field: Yes
Default Value: No default value
Column Name: BLG.BILL_PROCESSING_GRP

Section 4

Billing Rate Table (BLRT)

The Billing Rate Table (BLRT) defines the name of the billing rates used for each billing type and identifies the period of time the billing rate is effective. For example, an annual billing rate review requires one BLRT row per year for each billing rate in use and a quarterly billing rate review requires four BLRT rows per year for each billing rate in use. Remember that the end of a billing rate period does not require that you change the name of the billing rate. It just provides the option to change the name of the billing rate.

The key to the BLRT is a combination of the values in the FIID, Billing Type, and End Date fields. This key must be unique when a row is added and cannot be changed when a row is updated.

The following values must be defined in other tables before being used on BLRT screen 1:

- The value in the Billing Type field must first be defined in the Billing Type field on Billing Type Table (BLTY) screen 1.
- The value in the Rate Name field must first be defined in the Rate Name field on Rate Table (RATE) screen 1.

The value in the Billing Type field on BLRT screen 1 is used on file maintenance screens for several other tables. Therefore, a BLRT row should be available as long as that value is used in any of the following fields:

- The BILLING TYPE field on Bank Table (BANK) screen 1
- The BILLPAY BILLING TYPE field on Customer Table (CSTT) screen 1

The Billpay Table Maintenance Server process does not stop you from deleting a BLRT row that is being used in one of these tables. It is up to you to ensure that the BLRT row is no longer needed.

Screen 1

BLRT screen 1 contains the name of the billing rate used for each of the billing types for a specific FIID and effective date range. BLRT screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  BILLING RATE         LLLL       YY/MM/DD  HH:MM    1 OF  1

                 FIID ____

         Billing Type __

            Rate Name ________________

Effective Date Range:
           Begin Date ________ yyyymmdd
             End Date ________







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FIID — The financial institution identifier. The value in this field links the billing information in this row to a billing group and must match an FIID that already exists in the Bank Table (BANK), which is accessed from an Institution Definition File (IDF) screen. A row with values that match the values in this field and the Billing Type field must already exist in the Billing Type Table (BLTY).

Field Length:	1–4 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLRT.FIID

Billing Type — A code that identifies this group of customers or financial institutions for billing purposes throughout the BASE24-billpay database. A row with values that match the values in this field and the FIID field must already exist in the BLTY.

Refer to the “Suggested Naming Conventions” topic in section 1 for assigning a billing type.

Field Length: 1–2 alphanumeric characters
Required Field: Yes
Default Value: No default value
Column Name: BLRT.BILL_TYP

Rate Name — An identifier for the Rate Table (RATE) row that contains necessary rate information. This value must already exist in the RATE.

Refer to the “Suggested Naming Conventions” topic in section 1 for assigning rate names.

Field Length: 1–16 alphanumeric characters
Required Field: Yes
Default Value: No default value
Column Name: BLRT.RATE

Effective Date Range

The dates during which the billing rate named in this row is applicable. Rows for the same FIID and billing type cannot have effective date ranges that overlap. The ending date for this range should be the date of the next scheduled review of customer pricing. At that time, another BLRT row can be defined with a new rate name if pricing changes or the same rate name if pricing stays the same.

The format used to enter and display the dates in these fields is shown to the right of the Begin Date field and is controlled by the BP-DATE-FORMAT param in the Logical Network Configuration File (LCONF).

Begin Date — The first date when this row is applicable.

If the field is left blank, the system fills in the current date when the row is added or updated. However, the correct start date should be used so that effective date ranges do not overlap.

Field Length: 8 numeric characters
Required Field: No
Default Value: No default value
Column Name: BLRT.EFF_BEG_DATE

End Date — The last date when this row is applicable.

This field should contain the date of the next scheduled review of customer pricing because this date cannot be changed without deleting the BLRT row and replacing it with a new one. A new rate name can be defined if pricing changes or another BLRT row can be created with the same rate name if pricing stays the same.

If the field is left blank when the row is added or updated, the system assigns a date of December 31, 3999, but does not display it.

Field Length:	8 numeric characters
Required Field:	No
Default Value:	No default value
Column Name:	BLRT.EFF_END_DATE

Section 5

Billing Type Table (BLTY)

The Billing Type Table (BLTY) defines the groups of customers or financial institutions used for applying billing rates and cycles. The BLTY contains one row for each FIID and billing type combination.

An example of customer groups is a financial institution that has three groups of customers—preferred, regular, and commercial—and the preferred customers receive an introductory rate for the first three months before being switched to the regular rate. This financial institution would require rows in the BLTY for the following rates:

- Introductory rate for preferred customers
- Regular rate for preferred customers
- Rate for regular customers
- Rate for commercial customers

An example of a financial institution group is a billpay service provider that charges a set fee for each transaction occurring during the length of the service contract. This service provider would require only one row in the BLTY for the financial institution.

The key to the BLTY is a combination of the values in the FIID and Billing Type fields. This key must be unique when a row is added and cannot be changed when a row is updated.

The value in the Billing Type field on BLTY screen 1 is used on file maintenance screens for several other tables. Therefore, a BLTY row should be available as long as that value is used in any of the following fields:

- The BILLPAY BILLING TYPE field on Customer Table (CSTT) screen 1
- The BILLING TYPE field on Bank Table (BANK) screen 1

- The Billing Type field on Billing Rate Table (BLRT) screen 1
- The Next Billing Type field on another BLTY screen 1

The Billpay Table Maintenance Server process does not stop you from deleting a BLTY row that is being used in one of these tables. It is up to you to ensure that the BLTY row is no longer needed.

Screen 1

BLTY screen 1 contains general information for a billing type. BLTY screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  BILLING TYPE          LLLL        YY/MM/DD  HH:MM    1 OF  1

             FIID ____

      Billing Type __

       Description ______________________________

     Billing Usage _   (B=Bank, C=Customer)

  Number of Cycles ____

Next Billing Type __




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```

FIID — The financial institution identifier. The value in this field links the billing information in this row to a billing group and must match an FIID that already exists in the Bank Table (BANK), which is accessed from an Institution Definition File (IDF) screen.

Field Length:	1–4 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLTY.FIID

Billing Type — A code that identifies this group of customers or financial institutions for billing purposes throughout the BASE24-billpay database.

Refer to the “Suggested Naming Conventions” topic in section 1 for assigning billing type codes.

Field Length:	1–2 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	BLTY.BILL_TYP

Description — A free-format description of the billing type.

Field Length:	1–30 alphanumeric characters
Required Field:	No
Default Value:	No default value
Column Name:	BLTY.BILL_DESC

Billing Usage — A code identifying whether this row applies to a customer group or a financial institution group. Valid values are as follows:

B = Bank billing. The Bank Billing program uses this row to calculate billing amounts for a financial institution.
C = Customer billing. The Customer Billing program uses this row to calculate billing amounts for a customer.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	No default value
Column Name:	BLTY.BILL_USAGE

Number of Cycles — The number of billing cycles that this row is to be used for billing a customer or financial institution. Valid values are 1 through 9999.

A specific value is not reserved for identifying rows that are to be used indefinitely. However, entering the maximum value can serve this purpose since a value of 9999 means a row is to be used for 192 years with weekly billing cycles (that is, 52 billing cycles per year), and even longer periods of time if there are fewer billing cycles per year.

After the number of billing cycles specified in this field have been billed, the Customer Billing program uses the BLTY row specified in the Next Billing Type field for billing a customer. If the Next Billing Type field is blank or contains a value that is not defined in the Billing Type field of another BLTY row, no further billing amounts will be calculated.

The Bank Billing program does not use this value in processing. A valid entry is required, but is used for information only.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: No default value
Column Name: BLTY.NUM_CYCLES

Next Billing Type — The BLTY row to be used for billing a customer once the customer has been billed the number of cycles specified in this row. This field can contain blanks or the value defined in the Billing Type field of an existing BLTY row. If this field is blank or contains a value that is not defined in the Billing Type field of another BLTY row, the Customer Billing program calculates billing amounts only for the number of cycles specified in the Number Of Cycles field.

The Bank Billing program does not use this value in processing.

Field Length: 1–2 alphanumeric characters
Required Field: No
Default Value: No default value
Column Name: BLTY.NEXT_BILL_TYP

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Section 6

Rate Table (RATE)

The Rate Table (RATE) contains the information needed by the BASE24-billpay Billing Application to identify the BASE24-billpay transactions for which a financial institution or customer is charged. Each row contains a name for the rate, a combination of transaction source, transaction type, and action code to be considered, and the rate group used to calculate the service charge.

Since each combination of transaction source, transaction type, and action code requires a separate RATE row, multiple RATE rows may be needed to define one rate name. Also, there is no limit to the number of RATE rows that can use the same rate group.

The key to the RATE is a combination of the values in the Rate Name, Transaction Source, Transaction Type, and Action Code fields. This key must be unique when a row is added and cannot be changed when a row is updated.

The value in the Rate Group field on RATE screen 1 must first be defined in the Rate Group field on Rate Group Table (RATG) screen 1.

The value in the Rate Name field on RATE screen 1 is used in the Rate Name field on Billing Rate Table (BLRT) screen 1. Therefore, the RATE rows with a particular rate name should be available as long as the rate name is used in the BLRT.

Screen 1

RATE screen 1 identifies the rate group that applies to a set of transaction information. RATE screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  RATE                  LLLL       YY/MM/DD  HH:MM    1 OF  1

         Rate Name ________________

Transaction Source __  (AD=Audio DH, IB=Inbound, PC=Personal Computer,
                        SP=Screen Phone, Others Allowed)

  Transaction Type __

       Action Code ___

        Rate Group ________________




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```

Rate Name — The name of the rate to which the transaction identification information in this row belongs. Each rate name can have one or more rows in the RATE. However, this field cannot be blank and the combination of entries in all fields on this screen except the Rate Group field must be unique.

Refer to the “Suggested Naming Conventions” topic in section 1 for assigning a rate name.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATE.RATE

Transaction Source — A code identifying the type of device from which the transaction originates. The code can be a reserved value or be user-defined. The reserved values are as follows:

AD = Audio device (for example, interactive voice response system)
BL = Billing application
IB = Inbound (customer service representative terminal)
PC = Personal computer
SP = Screen phone

If a user-defined value is placed in this field, then some originating device or software must generate the user-defined value for its transactions and they must be logged to the History Table (HIST); otherwise, no service charges will be assessed based on this row.

Field Length:	2 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATE.TXN_SRC

Transaction Type — A code that identifies the transaction type. The code can be a reserved value or be user-defined. The reserved values are as follows:

3H = Inquire customer vendor list
3J = Inquire scheduled payments list
3K = Inquire scheduled transfers list
3L = Inquire last payments/transfers list
3M = Inquire scheduled transaction list
4A = Do scheduled transfer future
4B = Do scheduled transfer recurring
5A = Do scheduled payment future
5B = Do scheduled payment recurring
9A = Schedule payment
9B = Schedule payment future
9C = Schedule payment recurring
9D = Schedule transfer future
9E = Schedule transfer recurring
9F = Delete scheduled payment
9G = Delete scheduled transfer
9H = Update scheduled payment
9J = Update scheduled transfer
9K = Add/activate customer signup
9L = Inquire customer information
9M = Update customer information

9N = Add customer vendor
9P = Delete customer vendor
9Q = Update customer vendor
9R = Inquire master vendor list
9S = Add master vendor
9T = Request customer vendor list
9V = Add loan application
9W = Add miscellaneous request
9X = Inquire billpay history
B5 = Add/update customer signup accounts

Field Length:	2 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATE.TXN_TYP

Action Code — A code used by the transaction authorizer to indicate the processing outcome of a transaction. Valid values are as follows:

000 = Approved
001 = Honor with identification
002 = Approved for partial amount
003 = Approved (VIP)
004 = Approved, update Track 3
080 = Approved with backup account
081 = Approved with overdraft
100 = Deny, do not honor
101 = Expired card
102 = Suspected fraud
103 = Card acceptor contact acquirer
104 = Restricted card
105 = Card acceptor call acquirer's security department
106 = Deny, PIN tries exceeded
107 = Refer to card issuer
108 = Refer to card issuer, special conditions
109 = Invalid merchant
110 = Deny, amount invalid
111 = Deny, card number/ID invalid
112 = Deny, PIN is required
113 = Unacceptable fee
114 = Deny, no account of type
115 = Requested function not supported
116 = Deny, not sufficient funds
117 = Deny, incorrect PIN

118 = Deny, card/ID not found
119 = Transaction not permitted to cardholder
120 = Deny, txn not allowed at term
121 = Deny, wdl amt limit exceeded
122 = Deny, security violation
123 = Deny, wdl freq limit exceeded
124 = Violation of law
125 = Deny, card/ID not effective
127 = Deny, PIN length error
128 = Deny, PIN key synch error
180 = Deny, amount not found
181 = Deny, PIN change required
182 = Deny, new PIN invalid
183 = Deny, bank not found
184 = Deny, bank not effective
185 = Deny, customer vendor not found
186 = Deny, customer vendor not effective
187 = Deny, customer vendor account invalid
188 = Deny, vendor not found
189 = Deny, vendor not effective
190 = Deny, vendor data invalid
191 = Deny, payment date invalid
192 = Deny, personal ID not found
193 = Deny, scheduled transactions exist
200 = Do not honor
201 = Expired card
202 = Suspected fraud
203 = Card acceptor contact acquirer
204 = Restricted card
205 = Card acceptor call acquirer's security department
206 = Allowable PIN tries exceeded
207 = Special conditions
208 = Lost card
209 = Stolen card
300 = Successful
301 = Not supported by receiver
302 = Unable to locate record on file
303 = Duplicate record, old record replaced
304 = Field edit error
305 = File locked out
306 = Not successful
307 = Format error
400 = Accepted
500 = Accepted

600 = Accepted
800 = NMM OK
901 = Advice acknowledged, no financial liability accepted
902 = Deny, transaction invalid
903 = Reenter transaction
905 = Acquirer not supported by switch
906 = Cutover in process
907 = Issuer or switch inoperative
908 = Deny, route dest not found
909 = Deny, system malfunction
912 = Deny, issuer unavailable
913 = Deny, duplicate transmission
914 = Not able to trace back to original txn
915 = Reconciliation cutover or checkpoint error
919 = Key sync error
921 = Deny, security error
940 = Deny, database error
941 = Deny, invalid currency code
942 = Deny, amount format error
943 = Deny, customer vendor format error
944 = Deny, date format error
945 = Deny, name format error
946 = Deny, account format error
947 = Deny, recurring data error
948 = Deny, update not allowed

While this field accepts one, two, or three alphanumeric characters, BASE24 action codes are three alphanumeric characters in length. Use leading zeros in the field when necessary to match BASE24 action codes.

A description appears next to the Action Code field.

Field Length:	1–3 alphanumeric characters
Required Field:	No
Default Value:	No default value
Column Name:	RATE.RSP_CODE

Rate Group — A code identifying the group to which the rate information used with this RATE row belongs. This value must already exist in the RATG.

Refer to the “Suggested Naming Conventions” topic in section 1 for naming rate groups.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATE.RATE_GRP

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Section 7

Rate Group Table (RATG)

The Rate Group Table (RATG) contains details on how to calculate the fee assessed for a transaction. A rate group can have one or more RATG rows. Each row in the RATG identifies the rate group to which the rate belongs, the number of transactions to which the rate applies, the rate assessed for the transactions, and how the rate is used to calculate the total fee.

Two file maintenance screens—RATG screen 1 and the Rate Group List screen—are available with the RATG. RATG screen 1 lets you maintain individual rows of the RATG. The Rate Group List screen is an inquiry screen that lets you see all RATG rows in a rate group, arranged by values in the End Of Range field. Using the Rate Group List screen, you can change to another rate group or select a specific RATG row and return to RATG screen 1 to update that row.

The key to the RATG is a combination of the values in the Rate Group and End Of Range fields. This key must be unique and cannot be changed when a row is updated.

The value in the Rate Group field on RATG screen 1 is used in the Rate Group field on Rate Table (RATE) screen 1. Therefore, the RATG rows in a rate group should be available as long as the rate group is used in the RATE.

The Billpay Table Maintenance Server process does not stop you from deleting a RATG row that is being used in the RATE. It is up to you to ensure that the RATG row is no longer needed.

Calculating Fees

The RATG provides the following parameters required to calculate the fees assessed for using the BASE24-billpay product during a billing period.

- Transaction ranges
- Transaction group size
- Transaction group cost
- Cumulative indicator

Each RATG row contains one set of these parameters. A rate group can use one or more RATG rows.

The number of rate groups also affects fee calculations.

Examples of fee calculations using one rate group and multiple rate groups follow the parameter descriptions.

Parameter Descriptions

The following paragraphs provide detailed descriptions of the fee calculation parameters.

Transaction Ranges

The number of transactions during a billing period to which a rate applies is defined as a range (for example, 1 through 100 and 101 through 200). However, only the end of the range is specified in each RATG row (for example, 100 and 200). The rows in a rate group are arranged in ascending order by the range ending values (for example, 100 comes before 200). The start of each range is computed by adding one to the end of the previous range (for example, the second range's starting number of 101 is computed by adding 1 to the first range's ending number of 100). A range that ends with a value of zero identifies a fixed charge that is assessed regardless of the number of transactions. The range ending number is specified in the End Of Range field on RATG screen 1.

Transaction Group Size

Fees can be assessed on individual transactions or on groups of transactions that occur during a billing period. For example, if each transaction group is defined with five transactions, the specified cost is charged one time regardless of whether one, two, three, four, or five transactions have occurred. The sixth transaction starts a new transaction group and the specified cost is charged a second time. The specified charge is charged a third time when transaction number 11 occurs.

If each transaction group is defined with one transaction, the specified cost is assessed for each transaction. A fixed charge has a transaction group with one transaction. The size of the transaction group is specified in the Groups Of N field on RATG screen 1.

Transaction Group Cost

The transaction group cost is the amount assessed for each full or partial group of transactions that occurs during the billing period. The cost per transaction can be calculated by dividing the transaction group cost by the transaction group size. The cost for each group is specified in the Cost For Range field on RATG screen 1.

Cumulative Indicator

Each transaction range in a rate group has its own group size and group cost. When a rate group has more than one transaction range and the number of transactions for a billing period exceeds the first range, the cumulative flag controls which transaction group size and group cost are used to calculate the service fee, as follows:

- If the cumulative flag is off, the fee assessed for each group of transactions is calculated using the cost and group size specified for the transaction range to which the group of transactions belongs. Calculation example 1 illustrates this cumulative flag setting.
- If the cumulative flag is on, the fee assessed for each group of transactions is calculated using only the cost and group size specified for the highest applicable transaction range. Calculation example 2 illustrates this cumulative flag setting.

Calculation Examples

The following examples illustrate the fee calculation process. Each example includes the rows needed in the RATG for the rate group and the procedure used to calculate different numbers of transactions. The first example illustrates the calculation process when the cumulative flag is turned off. The second example illustrates the calculation process when the cumulative flag is turned on. The third example illustrates what happens when the transaction group size is always 1. The third example also eliminates the fixed charge used in the first two examples.

Example 1

The first example uses the following RATG rows:

Entry	End of Range	Groups of N	Cost for Range	Cumulative
1	0	1	1.00	N
2	10	1	0.50	N
3	100	10	2.00	N

A customer with no billable transactions is assessed $1.00. Entry 1 is a fixed charge that is assessed regardless of account activity.

A customer with 5 transactions is assessed $3.50. The $1.00 charge in entry 1 is fixed and the $0.50 charge in entry 2 is assessed for each of the transactions ($0.50 x 5 = $2.50).

A customer with 10 transactions is assessed $6.00. The $1.00 charge in entry 1 is fixed and the $0.50 charge in entry 2 is assessed for each of the transactions ($0.50 x 10 = $5.00).

A customer with 11 transactions is assessed $8.00. The $1.00 charge in entry 1 is fixed, the $0.50 charge in entry 2 is assessed for each of the first 10 transactions ($0.50 x 10 = $5.00), and the $2.00 charge in entry 3 is assessed for each additional group of up to 10 transactions.

A customer with as many as 20 transactions is also assessed $8.00 because no additional charge is assessed for transactions 12 through 20.

Example 2

The second example uses the following RATG rows:

Entry	End of Range	Groups of N	Cost for Range	Cumulative
1	0	1	1.00	Y
2	10	1	0.50	Y
3	100	10	2.00	Y

A customer with no billable transactions is assessed $1.00. Entry 1 is a fixed charge that is assessed regardless of account activity.

A customer with 5 transactions is assessed $3.50. The $1.00 charge in entry 1 is fixed and the $0.50 charge in entry 2 is assessed for each transaction ($0.50 x 5 = $2.50).

A customer with 10 transactions is assessed $6.00. The $1.00 charge in entry 1 is fixed and the $0.50 charge in entry 2 is assessed for each transaction ($0.50 x 10 = $5.00).

A customer with 11 transactions is assessed $5.00. The $1.00 charge in entry 1 is fixed and the $2.00 charge in entry 3 is assessed twice—once for transactions 1 through 10 and once for transaction 11.

A customer with as many as 20 transactions is also assessed $5.00 because no additional charge is assessed for transactions 12 through 20.

A customer with 21 transactions is assessed $7.00. Transaction number 21 starts the next group of 10 transactions and an additional $2.00 charge is assessed.

Example 3

The third example uses the following RATG rows:

Entry	End of Range	Groups of N	Cost for Range	Cumulative
1	10	1	0.50	Y
2	100	1	0.20	Y

A customer with no billable transactions is not assessed a service charge. This rate group does not have an entry for a fixed charge.

A customer with 5 transactions is assessed $2.50. The $0.50 charge in entry 1 is assessed for each transaction ($0.50 x 5).

A customer with 10 transactions is assessed $5.00. The $0.50 charge in entry 1 is assessed for each transaction ($0.50 x 10).

A customer with 11 transactions is assessed $2.20. The $0.20 charge in entry 2 is assessed for each transaction ($0.20 x 11 = $2.20) because there are more than 10 transactions. Entry 2 is used for the first 10 transactions instead of entry 1.

A customer with 20 transactions is assessed $4.00. The $0.20 charge in entry 2 is assessed for each transaction ($0.20 x 20 = $4.00).

A customer with 21 transactions is assessed $4.20. Entry 2 is the charge that is assessed for each transaction.

Number of Rate Groups Used

The number of rate groups you use in the Rate Table (RATE) also affects the way fees can be calculated. The Customer Billing and Bank Billing programs calculate the fee for each rate group and then combine the results. The fee calculation examples presented earlier use individual rate groups. The following fee calculation example illustrates how the number of rate groups affects the fee charged.

In this example, a customer performed the following immediate payment transactions.

Transaction	Source	Disposition
1	AD*	Approved
2	AD	Approved
3	IB†	Approved
4	IB	Denied for insufficient funds
5	IB	Approved
6	AD	Approved
7	IB	Approved
8	PC‡	Approved
9	AD	Approved
10	AD	Approved
11	AD	Approved
12	AD	Approved

* Audio device (for example, interactive voice response system)

† Inbound (customer service representative terminal)

‡ Personal computer

Example 1

If all RATE entries for approved transactions use the same rate group without distinguishing for the transaction source, eleven transactions (all transactions except number 4) are used in a single fee calculation. If the RATG entry is $0.50 per transaction, the total fee is $5.50 ($0.50 x 11 = $5.50).

Example 2

If the RATE entries for approved transactions use a separate rate group for each transaction source, four different fee calculations are performed and the results are added together, as shown in the following table:

Source	Transaction Number	Count	RATG Amount	Fee
AD	1, 2, 6, 9, 10, 11, 12	7	$0.50	$3.50
IB	3, 5, 7	3	$0.75	$2.25
PC	8	1	$1.00	$1.00
Total				$6.75

Example 3

If any rate group includes a fixed charge, the fixed charge is included in the calculation. You must use care with fixed charges in multiple rate groups because the fixed charges from all rate groups used are included in the calculation. For example, if the rate groups for transaction sources AD and IB include $3.00 fixed charges, the fee increases by $6.00, as shown below.

Source	Transaction Number	Count	Fixed Charge	RATG Amount	Fee
AD	1, 2, 6, 9, 10, 11, 12	7	$3.00	$0.50	$6.50
IB	3, 5, 7	3	$3.00	$0.75	$5.25
PC	8	1	$0.00	$1.00	$1.00
Total					$12.75

Screen 1 Function Keys

The use of one function key on RATG screen 1 varies from the standard function keys explained in the ***BASE24 Base Files Maintenance Manual*** and ***BASE24 CRT Access Manual.*** The use of this function key is explained below.

The first column shows the BASE24 key. The second column describes the function that can be accomplished with this key on the RATG screen.

Key	Function
F14	**DISPLAY RATE GROUP LIST —** Displays the Rate Group List screen containing all RATG rows with the same value in the Rate Group field as the RATG row you are currently viewing.

Screen 1

RATG screen 1 contains details on how to calculate the fee assessed for a transaction. RATG screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  RATE GROUP           LLLL       YY/MM/DD  HH:MM    1 OF  1

    Rate Group ________________

  End of Range _________________

   Groups of N _________________

Cost for Range ____________________

    Cumulative N (Y/N)






RECORD LAST CHANGED:                   BY USER:      ,
*********************************** BASE24 ***********************************
NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12-HELP                  F14-DISPLAY RATE GROUP LIST
```

Rate Group — A code identifying the group to which the rate information in this row belongs. Each rate group can have one or more rows in the RATG. This field cannot be blank and the combination of entries in this field and the End Of Range field must be unique.

Refer to the “Suggested Naming Conventions” topic in section 1 for naming the rate group.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATG.RATE_GRP

End Of Range — The maximum number of transactions that can occur during a billing period and use the rate information in this row.

Blanks in this field indicate the rate information in this row is a fixed cost that is always assessed regardless of transaction activity. Blanks are carried internally as the value 0.

The combination of entries in this field and the Rate Group field must be unique.

Field Length:	1–17 numeric characters
Required Field:	No
Default Value:	No default value
Column Name:	RATG.END_RANGE

Groups Of N — The number of transactions in each group for calculating service charges. The first transaction in each group causes the amount in the Cost For Range field to be assessed.

If the End of Range field contains blanks, the value 1 in this field indicates the rate information in this row is a fixed cost that is assessed regardless of transaction activity. If the End of Range field contains a value, the value 1 in this field indicates the rate information in this row is assessed for each transaction. Zero is not a valid value.

Field Length:	1–17 numeric characters
Required Field:	Yes
Default Value:	No default value
Column Name:	RATG.GRPS_OF_N

Cost For Range — The amount, expressed in whole and fractional currency units, charged for the number of transactions in the Groups Of N field.

Field Length:	1–20 numeric characters (2 decimal places are assumed)
Required Field:	No
Default Value:	No default value
Column Name:	RATG.COST_FOR_RANGE

Cumulative — A code indicating whether to use the values in the Groups Of N and Cost For Range fields of this row to calculate service charges for transactions from smaller ranges. Valid values are as follows:

Y = Yes, use this row.
N = No, do not use this row.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Column Name:	RATG.CUMULATIVE

Rate Group List Screen Function Keys

The use of three function keys on the Rate Group List screen varies from the standard function keys explained in the ***BASE24 Base Files Maintenance Manual*** and ***BASE24 CRT Access Manual.*** The use of these function keys is explained below.

The first column shows the BASE24 keys. The second column describes the functions that can be accomplished with these keys on the Rate Group List screen.

Key	Function
F14	**SELECT AND RETURN —** Selects the row identified by the cursor position and returns to the Rate Group screen with the information from that row displayed.
Shift-F7	**PREVIOUS PAGE —** Displays the information from the previous 13 rows of the RATG.
Shift-F8	**NEXT PAGE —** Displays the information from the next 13 rows of the RATG.

Rate Group List Screen

The Rate Group List screen displays all RATG rows in a selected rate group, in ascending order by values in the End Of Range field. Each page of the Rate Group List screen can display up to 13 RATG rows. From this screen, you can change to another Rate Group or select a specific rate group row and return to RATG screen 1 to update the selected row. The Rate Group List screen is shown below, followed by descriptions of its fields.

```
BASE24-BILLPAY  RATE GROUP LIST       LLLL       YY/MM/DD  HH:MM    1 OF  1

Rate Group ________________
                                                                  Cumulative
          End of Range           Groups of N        Cost for Range     (Y/N)
                     0











*********************************** BASE24 ***********************************
NEW PAGE:         FILE DESTINATION:         NEW LOGICAL NETWORK ID:
SF7-PREV PAGE  SF8-NEXT PAGE        F12-HELP       F14-SELECT AND RETURN
```

Rate Group — A code identifying the group to which the rate information displayed on this screen belongs.

Field Length:	1–16 alphanumeric characters
Required Field:	No
Default Value:	No default value
Column Name:	RATG.RATE_GRP

End Of Range — The maximum number of transactions that can occur during a billing period and use the rate information in this row.

Field Length:	System-protected
Column Name:	RATG.END_RANGE

Groups Of N — The number of transactions in each group for calculating service charges.

If the End of Range field contains the value 0 or blanks, the value 1 in this field indicates the rate information in this row is a fixed cost that is assessed regardless of transaction activity. If the End of Range field contains a value other than 0, the value 1 in this field indicates the rate information in this row is assessed for each transaction. Zero is not a valid value.

Field Length: System-protected
Column Name: RATG.GRPS_OF_N

Cost for Range — The amount, expressed in whole and fractional currency units, charged for the number of transactions in the Groups Of N field.

Field Length: System-protected
Column Name: RATG.COST_FOR_RANGE

Cumulative — A code indicating whether to use the values in the Groups Of N and Cost For Range fields of this row to calculate service charges for transactions from smaller ranges. Valid values are as follows:

Y = Yes, use this row.
N = No, do not use this row.

Field Length: System-protected
Column Name: RATG.CUMULATIVE

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Appendix A

Pricing Configuration Example

The following example illustrates the procedure to use in setting up pricing information in the following BASE24-billpay Billing Application tables:

- Billing Group Table (BLG)
- Billing Rate Table (BLRT)
- Billing Type Table (BLTY)
- Rate Table (RATE)
- Rate Group Table (RATG)

The pricing information must be placed in all of these tables to establish the codes that are then used in the Customer Table (CSTT) for each customer and the Bank Table (BANK) for each financial institution.

Scenario

Second Bank (BK02) has decided to offer its customers an automated bill payment service. Second Bank has contracted with a service bureau for the service rather than operate a bill payment service itself. This scenario illustrates the configuration required for customer billing and financial institution billing. The BASE24-billpay Billing Application Customer Billing program supports billing of Second Bank's customers. The BASE24-billpay Billing Application Bank Day and Bank Billing programs provide the information needed for the service bureau to compute the charges owed to it by Second Bank for processing services.

Second Bank has decided that customers for its bill payment service fall into one of two groups: regular or preferred. Only immediate payments will be allowed at this time. Second Bank will allow its customers to make payments using their personal computer, a touch tone phone, or by talking with an operator provided by the service bureau. The specifics are as follows:

- Preferred customers will be charged a monthly service fee of $14.95, which will include the first 25 approved immediate payments originating from a personal computer. A $0.50 fee will be assessed for each PC-originated immediate payment beyond the included 25. A $0.75 fee will be assessed for each approved immediate payment transaction originating from a touch tone phone or from interaction with a live operator. If a transaction is denied, regardless of reason, there will be no charge to a preferred customer. As an incentive, each preferred customer who signs up for the service receives the first three months at a reduced rate—only the monthly service fee will be assessed regardless of the number or source of transactions performed. Preferred customers will be billed on the 25th of each month.
- Regular customers will be charged a monthly service fee of $9.95 which includes the first five approved immediate payments originating from any source. A $0.50 fee will be assessed for each additional transaction, regardless of source. A $5.00 fee will be assessed for each immediate payment transaction denied due to non-sufficient funds (NSF). Regular customers will be billed on the 15th of each month.

Second Bank has contracted with the service bureau to provide the billpay service. The service bureau has several pricing tiers, and Second Bank qualifies for their level 2 rates. In their contract, both parties have agreed that Second Bank will be charged $.35 for each and every immediate payment attempt, whether approved or denied, regardless of how it was originated. This rate is good for six months and the service bureau bills their customers at the close of the business day on the last day of the month, on a bi-monthly basis (for example, Jan-Feb, Mar-Apr).

Preferred Customer Configuration

The BASE24-billpay database would be configured as described below for preferred customers. This information should be configured in the order presented.

Billing Group

A total of 12 entries are made in the Billing Group Table (BLG) for preferred customers—one to define each monthly cycle during the first year, as follows:

FIID	Billing Group	Billing Usage	Begin Date	End Date	Processing Group
----	BG1C	C	19961226	19970125	PG1C
----	BG1C	C	19970126	19970225	PG1C
----	BG1C	C	19970226	19970325	PG1C
----	BG1C	C	19970326	19970425	PG1C
----	BG1C	C	19970426	19970525	PG1C
----	BG1C	C	19970526	19970625	PG1C
----	BG1C	C	19970626	19970725	PG1C
----	BG1C	C	19970726	19970825	PG1C
----	BG1C	C	19970826	19970925	PG1C
----	BG1C	C	19970926	19971025	PG1C
----	BG1C	C	19971026	19971125	PG1C
----	BG1C	C	19971126	19971225	PG1C

Billing Type

A Billing Type Table (BLTY) entry is needed to define each group of customers that is to be billed in the same manner. There are two groups of preferred customers: introductory and regular. With an introductory offer of three months, each new preferred customer will be registered as PI (preferred introductory) for three months and changed to PR (preferred regular) beginning with the fourth month. The first entry reflects the introductory offer and the second entry reflects the on-going billing type. The BLTY entries are as follows:

FIID	Billing Type	Description	Billing Usage	Number of Cycles	Next Billing Type
BK02	PI	Preferred-Introductory	C	3	PR
BK02	PR	Preferred-Regular	C	9999	

Rate Group

The Rate Group Table (RATG) contains entries to define the pricing information. Preferred customers receive the introductory offer for the service fee only—no individual transaction fees apply. The Rate Group Table (RATG) entry to support this offer is as follows:

Rate Group	End of Range	Groups of N	Cost for Range	Cumulative
BK02-PI-A	0	1	14.95	N

Once the three-month introductory offer for preferred customers has expired, the preferred regular rates will apply. The RATG entries to support the preferred regular fee assessment schedule are as follows:

Rate Group	End of Range	Groups of N	Cost for Range	Cumulative
BK02-PR-PC-9A-A	0	1	14.95	N
BK02-PR-PC-9A-A	25	1	0.00	N
BK02-PR-PC-9A-A	9999	1	0.50	N

Rate Group	End of Range	Groups of N	Cost for Range	Cumulative
BK02-PR-AD-9A-A	9999	1	0.75	N
BK02-PR-IB-9A-A	9999	1	0.75	N

The first entry supports the monthly service charge of $14.95. The second entry supports the inclusion of the 25 approved immediate payments from a personal computer. The last three entries are included because additional transactions can be submitted from a personal computer and transactions could also be originated from a live operator or a touch tone phone.

The values in the Rate Group field follow the naming conventions outlined in the "Suggested Naming Conventions" topic in section 1. The last two entries in this table could be combined into a single entry because transactions originating from touch tone phones and from a live operator are being assessed at the same rate, $0.75 per transaction. Should the assessments diverge in the future depending upon the transaction source, the single entry would need to be entered as shown above.

Rate Table

The Rate Table (RATE) entries identify which transactions will be included in the service charge calculation. Two rate groups must be defined because fees are assessed for individual transactions once the introductory offer expires, but not while the introductory offer is in effect. A RATE entry is required for each combination of transaction source, transaction type, and action code that can be assessed a service charge once the introductory offer expires.

The RATE entries for introductory offer transactions are as follows:

Rate Name	Tran Source	Tran Type	Action Code	Rate Group
BK02-PI	AD	9A	000	BK02-PI-A
BK02-PI	IB	9A	000	BK02-PI-A
BK02-PI	PC	9A	000	BK02-PI-A

Once the introductory offer has expired for a given customer, fees are assessed for individual transactions. Only immediate payments (transaction type 9A) that have been approved (action code 000) are chargeable for preferred customers. The source of the transaction is a third defining characteristic of a chargeable transaction profile. The RATE entries for post-introductory offer transactions are as follows:

Rate Name	Tran Source	Tran Type	Action Code	Rate Group
BK02-PR	AD	9A	000	BK02-PR-AD-9A-A
BK02-PR	IB	9A	000	BK02-PR-IB-9A-A
BK02-PR	PC	9A	000	BK02-PR-PC-9A-A

There are six entries in the RATE because two RATE entries are needed for each combination of transaction source, transaction type, and action code for which a charge can be assessed—one is used during the introductory period and one is used following the introductory period.

Billing Rate Table

The Billing Rate Table (BLRT) identifies how long each rate in the RATE is valid. If rates for preferred customers are going to be reviewed annually, the BLRT entries are as follows:

FIID	Billing Type	Rate Name	Begin Date	End Date
BK02	PI	BK02-PI	19961226	19971225
BK02	PR	BK02-PR	19961226	19971225

Regular Customer Configuration

The BASE24-billpay database would be configured as described below for regular customers. This information should be configured in the order presented.

Billing Group

A total of 12 entries are made in the Billing Group Table (BLG) for regular customers—one to define each monthly cycle during the first year, as follows:

FIID	Billing Group	Billing Usage	Begin Date	End Date	Processing Group
----	BG2C	C	19961216	19970115	PG1C
----	BG2C	C	19970116	19970215	PG1C
----	BG2C	C	19970216	19970315	PG1C
----	BG2C	C	19970316	19970415	PG1C
----	BG2C	C	19970416	19970515	PG1C
----	BG2C	C	19970516	19970615	PG1C
----	BG2C	C	19970616	19970715	PG1C
----	BG2C	C	19970716	19970815	PG1C
----	BG2C	C	19970816	19970915	PG1C
----	BG2C	C	19970916	19971015	PG1C
----	BG2C	C	19971016	19971115	PG1C
----	BG2C	C	19971116	19971215	PG1C

Billing Type

A Billing Type Table (BLTY) entry is needed to define each group of customers that is to be billed in the same manner. There is one group of regular customers because there are no special offers and no value in the Next Billing Type field. The BLTY entry is as follows:

FIID	Billing Type	Description	Billing Usage	Number of Cycles	Next Billing Type
BK02	RG	Regular Customer	C	9999	

Rate Group

The Rate Group Table (RATG) contains entries to define the pricing information. The RATG entries to support the regular fee assessment schedule are as follows:

Rate Group	End of Range	Groups of N	Cost for Range	Cumulative
BK02-RG-9A-A	0	1	9.95	N
BK02-RG-9A-A	5	1	0.00	N
BK02-RG-9A-A	9999	1	0.50	N
BK02-RG-9A-D	9999	1	5.00	N

The first entry supports the monthly service charge of $9.95. The second entry supports the inclusion in the monthly service charge of five approved immediate payments from any source. The third entry supports Second Bank’s pricing schedule for approved immediate payments in excess of five. The final entry supports the $5.00 charge for each denial due to non-sufficient funds for any immediate payment.

The values in the Rate Group field follow the naming conventions outlined in the “Suggested Naming Conventions” topic in section 1 when the transaction source does not affect the pricing.

Rate Table

The Rate Table (RATE) entries identify which transactions will be included in the service charge calculation.

Immediate payments (transaction type 9A) that have been approved (action code 000) or denied due to non-sufficient funds (action code 116) are chargeable for regular customers. The source of the transaction must be included in RATE entries because it is the third defining characteristic of a chargeable transaction profile, even though it does not affect the amount charged. The RATE entries are as follows:

Rate Name	Tran Source	Tran Type	Action Code	Rate Group
BK02-RG	AD	9A	000	BK02-RG-9A-A
BK02-RG	AD	9A	116	BK02-RG-9A-D
BK02-RG	IB	9A	000	BK02-RG-9A-A
BK02-RG	IB	9A	116	BK02-RG-9A-D
BK02-RG	PC	9A	000	BK02-RG-9A-A
BK02-RG	PC	9A	116	BK02-RG-9A-D

There are six entries in the RATE because a separate RATE entry is needed for each combination of transaction source, transaction type, and action code.

Billing Rate Table

The Billing Rate Table (BLRT) identifies how long each rate in the RATE is valid. If rates for regular customers are going to be reviewed annually, the BLRT entry is as follows:

FIID	Billing Type	Rate Name	Begin Date	End Date
BK02	RG	BK02-RG	19961216	19971215

Second Bank Configuration

The BASE24-billpay database would be configured as described below for Second Bank. This information should be configured in the order presented.

Billing Group

A total of three entries are made in the Billing Group Table (BLG) for Second Bank—one to define each billing cycle in the initial pricing agreement, as follows:

FIID	Billing Group	Billing Usage	Begin Date	End Date	Processing Group
BK02	BG1B	B	19970101	19970228	PG1B
BK02	BG1B	B	19970301	19970430	PG1B
BK02	BG1B	B	19970501	19970630	PG1B

Billing Type

A Billing Type Table (BLTY) entry is needed to define each group that is to be billed in the same manner. Since the FIID must match a value already established in the Institution Definition File (IDF), you should establish a separate group of BLTY entries for each financial institution being billed. The BLTY entry for Second Bank is as follows:

FIID	Billing Type	Description	Billing Usage	Number of Cycles	Next Billing Type
BK02	L2	Level 2 Bank	B	3*	*

* The Number of Cycles and Next Billing Type fields are not used in processing for financial institution billing. However, the Number of Cycles field can be used for memo purposes to record information such as the number of cycles a contract runs. In this case, an entry of 3 reflects a six-month pricing agreement since each cycle is two months in length.

Rate Group

The Rate Group Table (RATG) contains entries to define the pricing information. Level 2 financial institutions such as Second Bank pay a set charge for each immediate payment transaction, regardless of result or the transaction source. The Rate Group Table (RATG) entry to support this pricing is as follows:

Rate Group	End of Range	Groups of N	Cost for Range	Cumulative
BK02-L2	9999	1	0.35	N

An alternative method of defining the cost of the service is to have a separate table entry for each combination of transaction source, transaction type, and action code, for a total of six table entries. The way the table is constructed in this example is the simplest way to set up the table. However, if the pricing on any combination changes, this table, along with the Rate Group Table and Billing Rate Table, will need to be re-worked to reflect a new pricing model.

Rate Table

The Rate Table (RATE) entries identify which transactions will be included in the service charge calculation.

Immediate payments (transaction type 9A) that have been approved (action code 000) or denied due to non-sufficient funds (action code 116) are chargeable to Second Bank. The source of the transaction must be included in RATE entries because it is the third defining characteristic of a chargeable transaction profile, even though it does not affect the amount charged. The RATE entries are as follows:

Rate Name	Tran Source	Tran Type	Action Code	Rate Group
LEVEL-2	AD	9A	000	BK02-L2
LEVEL-2	AD	9A	116	BK02-L2
LEVEL-2	IB	9A	000	BK02-L2
LEVEL-2	IB	9A	116	BK02-L2

Rate Name	Tran Source	Tran Type	Action Code	Rate Group
LEVEL-2	PC	9A	000	BK02-L2
LEVEL-2	PC	9A	116	BK02-L2

There are six entries in the RATE even though there is only one rate group defined in the RATG because a separate RATE entry is needed for each combination of transaction source, transaction type, and action code.

Billing Rate Table

The Billing Rate Table (BLRT) identifies how long each rate in the RATE is valid. Since the contract between the service bureau and Second Bank is valid for six months, the BLRT entry is as follows:

FIID	Billing Type	Rate Name	Begin Date	End Date
BK02	L2	LEVEL-2	19970101	19970630

Summary

The preceding sample stepped through configuring two types of customers for a bank that has contracted with a third party to offer the billpay service. The chargeable transaction set was limited in order to fully explain how the different tables interact and what the different table entries would look like to provide the correct billing.

A production environment involves the same tables as the ones shown in the sample. However, the number of entries in some of the tables, specifically the Rate Group Table (RATG) and Rate Table (RATE), may increase significantly. Where the case study utilized immediate payment transactions only, the complete list of transactions available within the BASE24-billpay product must be considered in a production environment. There are currently more than 30 different transaction types in the BASE24-billpay system that can be logged in the History Table (HIST) and used to compute service charges. Refer to the Transaction Type field description in section 6 for a complete list.

You should also consider the potential number of BASE24-billpay action codes that can be generated. This study utilized only the approved and denied-NSF action codes; however, there are several approved and denied action codes which must be considered in a true production environment. Refer to the Action Code field description in section 6 for a complete list.

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